

FINAL EXAM**Fall 2024**

- This is a closed-book exam. Only a 8.5"x11" formula sheet is allowed.
- No electronic devices are allowed, except for the simple calculator we provide.
- You may not communicate with other students during the exam. You will also keep the exam contents confidential.
- There are **180 points in the exam**. You also have **180 minutes**.
- Do not be concerned if your answers are not whole numbers. Large numbers are avoided to lower the burden of calculation. Don't worry if you find the numbers unrealistic---you could make sense of these numbers in your head by picking the right units yourself (but no need to answer with these units).
- Please be as complete as possible. Write down the reasonings for your answer (unless instructed otherwise).
- Complete all **SHORT ANSWER** questions in one Blue Book, and all **LONG PROBLEMS** in a 2nd Blue Book. **Write your name and "SHORT" / "LONG" on the cover of your Blue Books.**
- Answers written **anywhere else, including on the question sheet** will **NOT** be graded.
- Please return the **Blue Books** AND **the question sheet** after the exam.
- A camera will be set in the classroom to monitor exam situations (Exam recordings will be deleted once final grades are posted unless required for an ongoing Honor Committee case).
- Faculty will circulate in each classroom throughout the exam to answer clarifying questions.

Section I: SHORT QUESTIONS (80 points)

Please be concise. None of the answers require more than a brief paragraph. All questions are worth 10 points each.

- a) (10 points) True/False
- Except for rare cases, the demand curve is always downward sloping.
(True, 2.5pts)
 - A firm's average cost curve describes its supply function.
(False, 2.5pts)
 - Membership pricing extracts all surplus from consumers and therefore is always optimal.
(False, 2.5pts)
 - The demand for a product is less elastic when there are more substitutes.
(False, 2.5pt)
- b) (10 points) A local brewery uses both machines (m) and labor (l) to make their product. Suppose that they hire a consultant to estimate their marginal return of investments (MPI's) for machines and labor. They find that their MPI for labor is 0.05 and their MPI for machine is 0.06. How should they adjust their plan?

Answer: Half credit for "MPIs' have to be the same", then full credit for "more machines, fewer labor".

- c) (10 points) Dyson's cordless vacuum costs \$599. The average nationwide sales volume is 10K vacuums per day. On Black Friday of 2023, the price fell to \$299, and the nationwide sales volume became 20K. What is (approximately) the demand elasticity of Dyson's cordless vacuum? Suppose that (i) Dyson is a monopoly; (ii) there is a positive constant marginal cost for producing each unit of cordless vacuum; and (iii) demand is more elastic when price is lower, is \$599 an optimal price? Is it too high or too low?

Answer:

(75%) Percentage change in price: $(299-599)/599 \sim -0.5$; percentage change in quantity: $(20K-10K)/10K = 0.5$. Elasticity $\sim -0.5/0.5 = -1$.

(Full credit if used different bases for percentage changes)

(25%) Demand elasticity at \$599 is approximately -1. Marginal cost > 0 means markup should be < 1 according to the IEPR. So price should be higher

- d) (10 points) Orange Theory offers two kinds of membership plans, the Basic plan (\$79/mo) and the Elite plan (119/mo). The Basic plan provides 4 classes a month whereas the Elite plan provides 8 classes a month. Suppose that there are two types of potential customers, beginners and hardcore trainers. The hardcore trainers have higher reservation prices for each class and demand more classes in a month than the beginners. Suppose that the hardcore trainers have a surplus of \$20 if they choose the Elite plan, and suppose that Orange Theory is pricing optimally, what would the hardcore trainers' surplus be if they

choose the Basic plan?

Answer: The hardcore trainers are the “high types”. If Orange Theory is pricing optimally, they will be indifferent between choosing the Elite plan and the Basic plan, so their surplus should be \$20.

- e) (10 points) In May 2020, a box of surgical mask costs \$12, and there were 100K boxes sold nationwide in a day on average. In May 2024, a box of surgical mask costs \$10, and there were 20K boxes sold nationwide in a day on average. Comparing the prices and quantities of surgical masks in 2020 and in 2024, higher prices are coupled with higher quantities. Does this mean that the demand for surgical masks is upward sloping? Why or why not?

Answer: The information given here is not sufficient to draw any conclusions regarding the slope of the demand. Because equilibrium prices and quantities in 2020 versus 2024 depend on both the demand and supply, as well as their levels. What is more likely is that demand shifts down from 2020 to 2024, causing both the equilibrium price and the equilibrium quantity to fall.

- f) (10 points) The Government of Japan imposes a very high import duty on copper products from Chile, Peru and Australia. Who do you think would be the gainers and losers in Japan from such legislation?

Producers of copper products in Japan would gain.
Consumers would pay higher prices and so would lose.
Government would get tax revenue.

- g) (10 points) The state of Connecticut is considering imposing a tax on carbon emissions. What kind of tax is it and what is its purpose? Under which conditions would the imposition of such a tax lead to an efficient outcome?

Pigouvian of externality tax.

Objective is to reduce carbon emissions by equalizing private costs and social costs.

Tax is only efficient if it is possible to correctly price the externality / quantify the social costs of carbon emissions.

- h) (10 points) In the guessing game we played in the class, each player guesses an integer between 1 and 100, with the target number being $2/3$ of the average of all guesses. The players whose guesses were closest to the target number were the winners, and each winner received a reward of \$10.

Now, let's make TWO changes:

- The target number is now **the average** of all the guesses (instead of $2/3$ of the average).
- If there is more than one winner, they will split the reward of \$10 equally.

Assume there are at least three players in the game, and that each player aims to maximize their own reward. What are the Nash Equilibria of this new guessing game?

There are many Nash equilibria: every player guessing the same number is a NE.
 [Give at most 4 points if they only point out one of the NE's, or point out any wrong NE.]

(Note: the second change, together with having at least three players, rules out the possibility that when there are an even number of players, half of them guess, say, x and the other half guesses $x+1$.)

Section II: LONG PROBLEMS (100 points)

Problem 1 (20 points)

From the second quarter of 2020 to the fourth quarter of 2022, the median home sales price went up by 48%. This question explores a possible explanation for the rising cost of single-family houses.

Suppose that, in 2020, the demand for houses was:

$$Q_D(p) = 150 - 100p$$

Meanwhile, the supply of houses in 2020 was:

$$Q_S(p) = 50p.$$

- a) (3 points) What is the equilibrium price and quantity of the housing market in 2020?

Price=1; quantity=50

In 2020, people started to spend a lot more time at home because of Covid. Therefore, many people wanted to scale up their houses. Moreover, the booming stock market gave a lot of cash to individual investors, causing them to have higher willingness to pay. This leads to an increase in demand. Suppose that the demand for houses in 2022 became:

$$Q_D(p) = 300 - 50p$$

- b) (3 points) What would the equilibrium price and quantity be if the supply had stayed the same?

Price=3; quantity=150

Moreover, the supply chain problem made it difficult and more costly for new constructions to gather materials. Social distancing and other Covid prevention measures also made any in-person work in the construction process (e.g., wiring, plumbing, and application of permits etc) difficult. These caused the supply to decrease. Suppose that the supply in 2022 became:

$$Q_S(p) = 10p.$$

- c) (4 points) What is the equilibrium price and quantity in 2022?

Price=5; quantity=50

To make housing more affordable, suppose that the government decides to provide subsidies to home buyers.

If there is a \$1 per-unit subsidy to home buyers, what would the equilibrium price and quantity be? What is the total expenditure? What is the deadweight loss?

- d) (5 points) If there is a \$1 per-unit subsidy to home buyers, what would the equilibrium price and quantity be? What is the total expenditure? What is the deadweight loss?

Inverse demand before subsidy:

$$P_D(q) = 6 - 0.02q$$

Inverse demand after subsidy:

$$P_D(q) = 7 - 0.02q$$

Inverse supply:

$$P_S^{\square}(q) = 0.1q$$

Equilibrium quantity: $0.12q=7$, $q^*=58.33$.

Equilibrium price paid by consumers: $6-0.02*58.33=4.83$.

Equilibrium price received by producers: 5.83.

Total subsidy= $1*58.33=58.33$.

Deadweight loss: $\frac{1}{2}*1*(58.33-50)=4.165$.

- e) (5 points) Would it be better for the producers if they were subsidized \$1 per unit instead of the consumers? Why or why not? Which side of the market benefits more from subsidies?

Subsidizing the producers would be the same as subsidizing the consumers. Demand elasticity in the 2022 equilibrium is $50*5/50=5$. Supply elasticity in the 2022 equilibrium is $10*5/50=1$. Demand is more elastic, so consumers always benefit more regardless which side is subsidized.

Problem 2 (24 points)

CalTrain has a local monopoly in the market for railway services between Palo Alto and San Francisco. The inverse demand for rides is given by $P(Q) = 12 - (2/25)Q$, where Q is the number of tickets sold and P is the price of a ride. CalTrain incurs in variable cost of \$2 per train ride and has fixed costs of zero.

- a) (6 points) Calculate the profit maximizing price and quantity for Caltrain.

Revenues are $P(Q) * Q = 12Q - 2/25Q^2$, so marginal revenues equal $12 - 4/25 Q$.

Marginal costs equal 2.

MR = MC when $Q=62.5$ and $P = 7$.

- b) (6 points) What are the company's profits? What is consumer surplus? What is the deadweight loss?

Profits are equal to $P*Q - FC - VC * Q = 7*62.5 - 2*62.5 = 312.5$.

The consumer surplus is the area between the demand curve and the monopoly price. The base equals monopoly demand, i.e., 62.5. The height is the difference between the intercept of the inverse demand curve (12) and the price (7), i.e., 5. The area is thus $5 \times 62.5 / 2 = 156.25$.

The deadweight loss can be found as the area of the triangle with height the difference between price and marginal cost ($7 - 2 = 5$) and base the difference in the monopoly quantity (62.5) and the quantity under perfect competition. To find the latter we can just equate demand to supply, i.e., marginal cost): $12 - 2/25Q = 2$, which implies $Q = 125$. Hence, the area of the deadweight loss triangle is $(125 - 62.5) \times 5 / 2 = 156.25$.

[Note: Give half credit to students who identify the areas graphically but do not calculate them.]

- c) (6 points) Suppose that, upon further market research, CalTrain finds out that they are serving two distinct groups of customers: Business, with an inverse demand of $P(Q) = 20 - 0.4Q$, and Students, with an inverse demand of $P(Q) = 10 - 0.1Q$. If CalTrain can price-discriminate, what would be the prices offered to the two groups of customers? What is the total number of tickets sold? What is the profit?

We can treat the two markets as two separate monopolistic markets.

For business, $MR = MC$ implies $Q = 22.5$ and $P = 11$. The profit is equal to 202.5.

For students, $MR = MC$ implies $Q = 40$ and $P = 6$. The profit is equal to 160.

With price discrimination, total profits equal $160 + 202.5 = 362.5$.

- d) (2 points) Given your answers to points a), b), and c), should CalTrain price-discriminate or not? Explain.

Yes, since profits are larger in d).

- e) (4 points) Suppose CalTrain is now able to engage in perfect price discrimination. Considering the total inverse demand of point (a), what would be the company's profits and what would be consumer surplus?

Under perfect price discrimination, consumer surplus is zero and profits correspond to the area of the triangle below the demand curve and above marginal cost. Since total quantity sold would be where demand equals supply, the base of the triangle is 125. The height of the triangle is the intercept between the intercept of the inverse demand curve (12) and marginal costs (2), i.e., 10.

Total profits would thus be $10 \cdot 125 / 2 = 625$. Note that this is also the sum of the deadweight loss, profit, and consumer surplus in point b).

Problem 3 (21 points)

A friend is deciding between two job offers: one from firm Flat, which pays a flat wage of \$95,000, and one from firm P4P, which pays a base salary of \$75,000 and a bonus of \$45,000 upon reaching a performance target. Suppose your friend assigns a 50% probability to reaching the performance target.

- a) (2 points) What is the expected value of the two jobs to your friend?

Flat: \$95,000

P4P: $\$75,000 + 0.5 \cdot \$45,000 = \$97,500$

- b) (2 points) If your friend is risk-neutral, what job would they pick?

They would pick the P4P job (utility = expected value, so they pick the job with the highest expected value).

- c) (3 points) Would your answer change if your friend is risk-averse, with a utility function $u(x) = \sqrt{x}$?

Now utility is not equal to expected value anymore.

Expected utility of Flat: $\sqrt{95,000} = 308.2207$

Expected utility of P2P: $0.5 \cdot \sqrt{75,000} + 0.5 \cdot \sqrt{120,000} = 310.1357$

The answer would not change; EU of P4P > EU of Flat.

Suppose now that you are a HR manager at firm P4P and are deciding whether to change your compensation scheme. Namely, you are considering adjusting your base salary and bonus to incentivize employees to work hard. Suppose your employees can choose two levels of effort: “low”, which gives them a chance equal to 30% of meeting the performance target, and “high”, which gives them a chance equal to 70%. The cost of effort is equal to \$30,000. Assume, as above, that your main competitor – firm Flat – is paying employees a flat wage of \$95,000.

- d) (6 points) Assuming all employees are risk-neutral, find the optimal base salary and bonus you would have to pay if you wanted to implement the bonus scheme.

The setup is identical to what we have seen in class, with $S=0.7$, $n=0.3$, $c=30,000$, and $m = 95,000$.

The optimal bonus is $b^* = c/(S-n) = 30,000/0.4 = \$75,000$

The optimal base salary is $w = m - n*b = 95,000 - 0.3*75,000 = \$72,500$

- a) (2 points) Would your answers to point d) change if you found out your employees were risk averse? Why?

The answer to point d) assumes that workers are risk averse. If workers are risk-neutral, they would have to be compensated for the risk involved in the bonus scheme, so at least one between w and b would have to be higher.

- b) (6 points) Suppose that, upon further research, you find out that the profit margins from meeting the performance target is \$60,000 per worker (that is, each worker meeting the target increases your profits, net of all non-labor costs, by \$60,000). Would it be worth it for your company to stick with the bonus scheme, or would it be better off to switch to the flat wage scheme? (For this point, assume again workers are risk-neutral). Explain.

We saw in class that the per-worker, per-unit of output (in this case, the performance target) profit margin would have to be at least as great as the bonus in order to justify the bonus scheme being used. Given that we found a bonus of \$75,000 in point d), a margin of \$60,000 is too low and P4P is better off offering the same flat wage as the competitor.

Game theory long questions (35 points)

Problem 4: Price Competition and Capacity [20 points]

Consider a one-shot price competition between Apple and Samsung for their new smartphones. Each company can choose to charge either \$800 or \$1,000. The following is the DEMAND matrix (not the profit matrix):

		Samsung	
		\$800	\$1000
Apple	\$800	30, 20	35, 10
	\$1000	18, 22	21, 18

For example, if both firms set a price of \$800, Apple will have a demand of 30 million units and Samsung will have a demand of 20 million units. (While the demand numbers are in Millions, you

don't need to account for that in your calculations.) Suppose **the production cost per unit is \$400**, the same for both companies. (We don't consider fixed costs as they usually don't impact price competition.) Assume each company aims to maximize its own profits.

a) Suppose first that the two companies set their prices simultaneously. Please answer the following questions. [Hint: Start by constructing the game matrix with profit numbers. Don't forget to include costs, and be sure to double check your calculations before proceeding.]

(a-1) Does Samsung have a dominant strategy? If so, please specify. **[4 points]**

(a-2) What prices will the two companies set in the Nash Equilibrium? **[4 points]**

		Samsung	
		\$800	\$1000
Apple	\$800	12k, 8k	14k, 6k
	\$1000	10.8k, 8.8k	12.6k, 10.8k

All the profit numbers are calculated as $\text{profit} = \text{demand} \times (\text{price} - \text{cost})$.

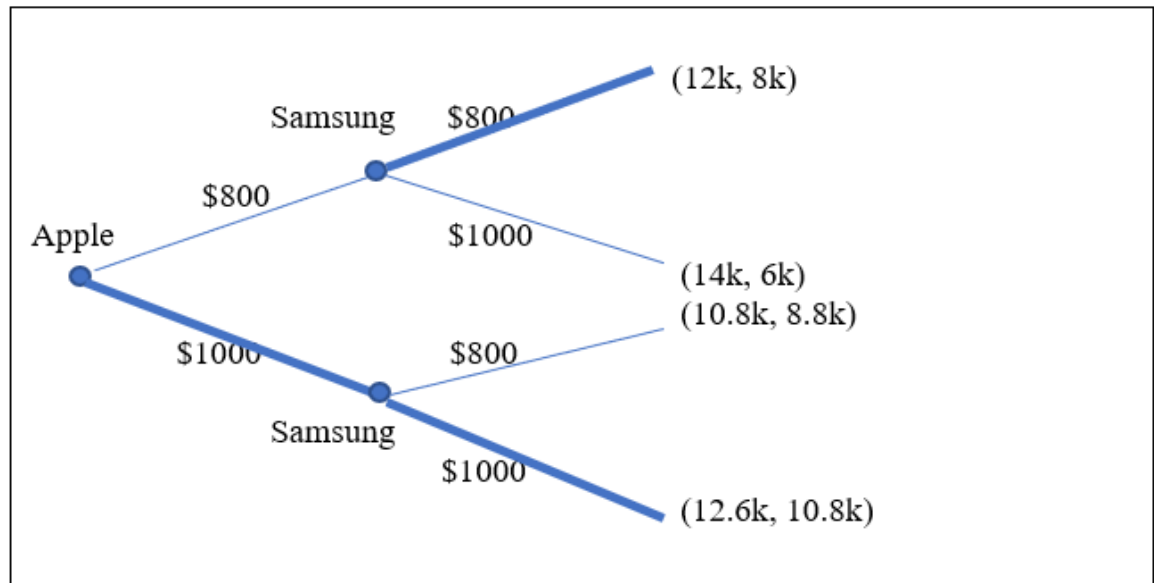
(a-1) Samsung has no dominant strategy. **[4 points]**

(a-2) In NE, each firm charges \$800. (This is easy to see by noticing that Apple has a dominant strategy \$800, or by the best response method.) **[2 points for each firm's price]**

b) Suppose now that Apple announces its price first, and then Samsung responds. Please answer the following questions. [Hint: Start by drawing the game tree.]

(b-1) What prices will the two companies set? **[4 points]**

(b-2) If instead Samsung announces its price first, how will each company's profit change? **[4 points]**



(b-1) Both firms will charge \$1000. [2 points for each firm's price]

(b-2) In that case, both firms will charge \$800. Apple's profit will drop from 12.6k to 12k, and Samsung's profit will drop from 10.8k to 8k. [2 points for each firm; it is ok if they just say each firm's profit drops.]

c) Return to the simultaneous-move game, but now suppose Apple has a capacity constraint of 25 million units, while Samsung does not. In other words, Apple cannot serve more than 25 million customers, and any unserved customers will buy from Samsung. Please answer the following questions. [Hint: You need to re-draw a game matrix.]

(c-1) What prices will the two companies now set in the Nash Equilibrium? [2 points]

(c-2) Compared to the situation in question (a), does Apple suffer or benefit from its capacity constraint, and by how much? [2 points]

		Samsung	
		\$800	\$1000
Apple	\$800	10k, 10k	10k, 12k
	\$1000	10.8k, 8.8k	12.6k, 10.8k

(c-1) Both charge \$1000. [1 point for each firm's price]

(c-2) Apple benefits from its capacity constraint by $0.6k$. [2 points] (This is because the capacity constraint softens price competition.)

Problem 5: A game with some unknown payoffs [15 points]

Consider the following simultaneous-move game with two unknown payoff numbers X and Y . $X \neq 3$, $Y \neq 7$, and they are integers.

		Player 2		
		L	C	R
Player 1	T	9, 5	X , 6	2, 4
	M	6, 5	2, 4	1, 3
	B	2, 7	3, 6	4, Y

(a) If $X=5$ and $Y=8$, does any player have a dominant or dominated strategy? [3 points]

Player 1 has a dominated strategy M.

(b) For what values of X and Y does the game have two Nash equilibria (NE)? [6 points]
[Hint: find all possible values of X and Y , and express your answers in inequalities.]

$X \geq 3$ and $Y \geq 7$ (or $X > 3$ and $Y > 7$). [3 points for each value.]

(c) For what values of X and Y does the game have a unique NE that predicts the same outcome as in the sequential-move version of this game, where player 1 moves first? [6 points]
[Hint: find all possible values of X and Y ; try to use backward induction **without** drawing the game tree for the sequential game.]

$X > 6$ and $Y < 7$. [3 points for each value] (Give full points if they use weak inequalities.)

There are two possible cases where the simultaneous game has a unique NE: $(X \geq 3, Y < 7)$ in which case the NE is (T, C), and $(X < 3, Y \geq 7)$ in which case the NE is (B, R).

In the sequential-move game:

if player 1 chooses T, then player 2 will choose C, so player 1 will get X ;

if player 1 chooses M, then player 2 will choose L, so player 1 will get 6;

if player 1 chooses B, then player 2 will choose L or R, depending on whether $Y < 7$ or $Y > 7$, so player 1 will get 2 or 4. In either case the outcome is worse for player 1 than 6 when she chooses M.

Therefore, player 1 will NEVER choose B. The outcome is either (T, C) if $X > 6$ or (M, L) if $X < 6$.

Therefore, to make the simultaneous and the sequential game have the same (unique) prediction, the only possibility is that (T, C) happens in either case.

For (T, C) to be the unique NE in the simultaneous game, as pointed out before, we need $(X \geq 3, Y < 7)$; for the same outcome to happen in the sequential game, we need $X > 6$. Therefore, the condition we are seeking is $(X > 6, Y < 7)$.